

Lappeenrannan teknillinen yliopisto
LUT School of Engineering Sciences

Software Development Skills

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LEARNING DIARY, <Software Development Skills: Mobile 2025-26 >

LEARNING DIARY – ANDROID MOBILE DEVELOPMENT

05.05.2026

Today I started the Android Mobile Development course and read through the course information carefully. I learned that the goal of the course is not only to learn Android development but also to create a unique project that demonstrates my skills as a software developer. I reviewed the course requirements and understood that I would need to complete tutorial exercises, maintain a learning diary, and create a final project.

I installed Android Studio and started exploring the development environment. Initially, there were many unfamiliar menus and settings, but I gradually became more comfortable navigating the interface. I also learned about the importance of version control and created a Git repository for the course work.

06.05.2026

Today I focused on setting up the Android development environment. I created my first Android Studio project and learned about the structure of an Android application. I explored folders such as resources, manifests, and source code directories.

I also configured an Android Emulator. At first, the emulator setup process was confusing because there were many device options and system images available. After some experimentation, I successfully created and launched a virtual Android device.

The biggest lesson today was understanding how Android Studio organizes projects and how applications are deployed to Android devices.

07.05.2026

Today I continued working on the Introduction module. I learned how Android Studio can be used to create a simple application. I modified text elements and experimented with changing user interface components.

I learned that Android applications consist of both visual elements and program logic. Even a simple application requires interaction between code and the user interface. Running the

application and immediately seeing changes appear on the emulator helped me understand the development process.

I also made several Git commits to save my progress.

08.05.2026

Today's focus was running and testing Android applications. I learned how to build a project and launch it through Android Studio. I experimented with making small changes and repeatedly running the application to verify the results.

I encountered some build errors and spent time reading error messages. This helped me understand that error messages often provide useful information about the source of a problem. Learning how to interpret these messages reduced my frustration and improved my confidence.

09.05.2026

Today I learned more about debugging Android applications. I explored Logcat and observed how Android Studio displays information while applications are running.

I intentionally introduced a few small mistakes into the code and observed how Android Studio reported the errors. This exercise helped me understand the importance of debugging during software development.

I learned that debugging is not only about fixing problems but also about understanding application behavior.

10.05.2026

Today I reviewed everything covered in the Introduction module. I practiced creating new projects, modifying interface elements, and running applications through the emulator.

I became more familiar with the Android Studio environment and felt much more comfortable navigating files and project structures. Compared to the first day, many concepts that initially seemed complicated now felt easier to understand.

11.05.2026

I completed the Introduction Part 1 exercises and reflected on what I had learned during the first week.

The most important lessons were:

- Setting up Android Studio.
- Creating Android projects.
- Running applications on an emulator.
- Debugging applications.
- Understanding Android project structure.
- Using Git for version control.

By the end of the week I felt ready to move on to more advanced Android concepts.

12.05.2026

Today I started the Core Elements module. I learned about the fundamental building blocks of Android applications.

The most important concept introduced was Activities. I learned that activities represent screens in Android applications and are responsible for displaying user interfaces and handling user interaction.

Understanding activities helped me see how larger Android applications are structured.

13.05.2026

Today I focused on learning about Intents. I learned that Intents allow Android applications to communicate between activities and move users from one screen to another.

I practiced examples where information was passed between activities. This helped me understand how applications can maintain workflows across multiple screens.

I realized that Intents would later be very useful in my own project because my application would require navigation between several pages.

14.05.2026

Today I continued studying Intents and application navigation.

I experimented with different examples and became more comfortable with how Android applications manage transitions between screens. I learned that navigation is one of the most important aspects of mobile application development because users expect smooth movement throughout the application.

15.05.2026

Today's topic was Intent Services. I learned that some application tasks can be executed in the background without blocking the user interface.

Although modern Android development often uses newer approaches, studying Intent Services helped me understand the concept of background processing and why it is important for maintaining a responsive application.

16.05.2026

Today I studied Broadcast Receivers. I learned that Android applications can listen for system events such as battery status changes, network connectivity changes, and notifications.

I found this concept interesting because it demonstrated how applications can react to events happening outside the application itself.

17.05.2026

Today I reviewed all concepts from the Core Elements module. I compared Activities, Intents, Intent Services, and Broadcast Receivers and reflected on their different responsibilities within Android applications.

I also began thinking about ideas for my final project and how navigation between multiple screens could be implemented.

18.05.2026

I completed the Core Elements module and summarized my learning.

The most important concepts learned during this period were:

- Activities.
- Intents.
- Navigation between screens.
- Background processing.
- Broadcast Receivers.
- Android application architecture.

These concepts provided a strong foundation for building larger Android applications and would later become important when developing my own project.

19.05.2026

Today I started the Lists, Layouts, and Images module. I learned about ListView and how it can be used to display multiple pieces of information in a structured way. I experimented with displaying data in list format and observed how Android applications organize large amounts of content.

I realized that displaying information clearly is important because users should be able to find information quickly without becoming overwhelmed.

20.05.2026

Today I continued learning about ListView and information presentation. I explored how lists can be customized and how different types of information can be displayed within the same application.

I learned that reusable components are important because they reduce duplicated work and make applications easier to maintain.

21.05.2026

Today's focus was Android layouts. I learned how layouts determine the structure and positioning of user interface elements on the screen.

I experimented with arranging text, buttons, and images in different ways. Through practice, I learned that good layouts improve usability and create a more professional appearance.

I also discovered that spacing and alignment play a major role in user experience.

22.05.2026

Today I studied custom layout components. I learned that developers can create reusable user interface elements that can be used repeatedly throughout an application.

This concept interested me because reusable components improve consistency and reduce the amount of code needed in larger projects.

I started thinking about how I could use custom components in my own Android project later.

23.05.2026

Today's topic was working with images in Android applications. I learned how images are stored in drawable resources and how they can be displayed using ImageView.

I experimented with different image sizes and observed how images affect the visual appearance of an application.

I realized that images are especially important in tourism applications because they help users visualize destinations and attractions.

24.05.2026

Today I continued working with images and layouts. I practiced combining text and images into a single interface.

I learned that visual design is not only about appearance but also about communication.

Images help users understand information more quickly than text alone.

This inspired me to create a project that would make extensive use of images.

25.05.2026

Today I completed the Lists, Layouts, and Images module.

The most important concepts I learned were:

- Displaying information using lists.
- Creating layouts.
- Designing reusable components.
- Working with images.
- Organizing user interfaces effectively.

These skills would become essential during the development of my final project.

26.05.2026

Today I started planning my final Android project. I wanted to create something that demonstrated the concepts learned during the course while also reflecting my personal interests.

After considering several ideas, I decided to create a tourism application called "Discover Gran Canaria."

The goal of the application would be to provide information about the island's nature, culture, beaches, food, wildlife, cities, and sustainability efforts.

27.05.2026

Today I focused on project planning and structure.

I created an outline of the application's screens and navigation flow. I identified the main pages that would be needed and considered how users would move between different sections.

I also researched information about Gran Canaria and collected ideas for content that could be included in the application.

28.05.2026

Today I continued gathering resources for the project.

I searched for images that could be used to represent beaches, mountains, cities, wildlife, plants, and cultural attractions. I also collected information that would later be displayed within the application.

This preparation helped me understand the importance of content collection before implementation begins.

29.05.2026

Today I started creating the initial Android project structure for Discover Gran Canaria.

I created a new Android Studio project and began setting up navigation. I also planned how different screens would connect to each other.

I spent time thinking about the user experience and how to make navigation intuitive.

30.05.2026

Today I worked on the visual identity of the project.

I considered color schemes, image styles, and overall design choices that would match the tourism theme. I wanted the application to feel welcoming and visually appealing.

I learned that design decisions have a significant impact on how users perceive an application.

31.05.2026

Today I reviewed the project structure and adjusted some of my original plans.

As I learned more about Android development, I identified opportunities to make the application more organized and easier to expand in the future.

This process taught me that software projects often evolve during development.

01.06.2026

Today I finalized the planning phase and prepared to begin full development of the Discover Gran Canaria application.

I reviewed the skills learned throughout the tutorial modules and considered how they would be applied in my own project.

At this stage I had a clear vision of the application structure and was ready to begin implementing the actual functionality and user interface.

02.06.2026

Today I began developing the Discover Gran Canaria application. I started by creating the main project structure and setting up navigation. I reviewed the Android Studio project files and organized the application so that new screens could be added easily.

I also started building the home screen and experimented with different layouts and image placements. My goal was to create an attractive first impression for users opening the application.

03.06.2026

Today I focused on designing the home page. I added a title, descriptive text, and a button that would later navigate users to the main content of the application.

I learned more about spacing, alignment, and text styling in Jetpack Compose. Small design adjustments made a significant difference to the appearance of the application.

I also tested the application repeatedly to ensure that all components displayed correctly.

04.06.2026

Today I worked on navigation between screens. I learned how navigation components can be used to move users from one page to another.

I created routes for multiple screens and practiced connecting them together. Initially, some navigation paths did not work correctly, but after reviewing the code and testing different solutions, I successfully connected the screens.

This helped me understand the importance of application flow and user experience.

05.06.2026

Today I created the Explore screen, which serves as the central navigation hub of the application.

I designed category cards that allow users to access different topics related to Gran Canaria. I began implementing sections for nature, beaches, culture, food, wildlife, plants, cities, and sustainability.

I learned how reusable components can reduce duplicated code and improve consistency across the application.

06.06.2026

Today I focused on the Nature section. I researched attractions such as Roque Nublo, Pico de las Nieves, and Tamadaba Natural Park.

I created content pages containing information and images related to these natural attractions. This required organizing both textual and visual information in a clear and attractive way.

I also improved my understanding of image handling within Android applications.

07.06.2026

Today I continued developing the Nature section and improved navigation between different nature-related pages.

I implemented buttons and navigation routes that allow users to move between attractions. Testing these features helped me understand how larger Android applications manage multiple screens.

I also improved the visual design of the pages by adjusting spacing and image placement.

08.06.2026

Today I worked on the Beaches section of the application.

I added information about famous beaches in Gran Canaria and incorporated images to make the content more engaging. I learned that visual content is especially important in tourism applications because users often make decisions based on what they see.

I also reviewed previous screens and corrected minor layout issues.

09.06.2026

Today I focused on the Culture section. I researched historical locations and cultural attractions in Gran Canaria.

I created pages that present information in a structured and easy-to-read format. I experimented with different arrangements of text and images to improve readability.

I learned that balancing information and visual content is important for maintaining user interest.

10.06.2026

Today I worked on the Food section.

I researched traditional dishes from Gran Canaria and incorporated information about local cuisine into the application.

I continued using reusable layout components to keep the design consistent across all sections of the application.

This process helped me better understand component-based design.

11.06.2026

Today I focused on Wildlife and Plants sections.

I added information about local animal species, marine life, and unique plant species found on the island. I also included images to support the information.

This stage reinforced the importance of organizing content effectively within an application.

12.06.2026

Today I developed the Cities section.

I added information about Las Palmas and other important urban areas. I wanted users to learn not only about nature but also about the people and communities living on the island.

I also reviewed navigation and made improvements to the overall application structure.

13.06.2026

Today I worked on the Sustainability section.

I researched environmental initiatives and sustainability efforts taking place in Gran Canaria. This section helped make the application more educational by highlighting environmental awareness and conservation.

I learned how software projects can be used not only for entertainment but also for sharing useful information.

14.06.2026

Today I reviewed all screens created so far.

I checked for consistency in layouts, fonts, images, and navigation. Several minor improvements were made to spacing and alignment.

Testing revealed a few navigation issues, which I corrected after reviewing the code.

15.06.2026

Today I improved the visual appearance of the application.

I adjusted image sizes, card layouts, and overall design. I wanted the application to feel more professional and polished.

I spent considerable time comparing different layout options and selecting those that provided the best user experience.

16.06.2026

Today I implemented image slideshows for the application.

I learned how Kotlin coroutines and `LaunchedEffect` can be used to update images automatically over time. This feature made the application feel more dynamic and interactive.

Initially, I encountered difficulties with image transitions, but after testing and debugging I successfully implemented the slideshow functionality.

‘17.06.2026

Today I continued refining the slideshow feature and improved overall navigation.

I tested the application repeatedly to ensure that all routes worked correctly and that users could move smoothly between screens.

I also cleaned up portions of the code to improve readability and maintainability.

18.06.2026

Today I added a Frequently Asked Questions section to the home page.

The FAQ section allows users to interact with questions and reveal answers when needed. This introduced an additional interactive component into the application.

I learned how state management can be used to control user interactions within Jetpack Compose.

19.06.2026

Today I focused on testing and debugging.

I reviewed every screen in the application and ensured that navigation, images, and interactive components functioned correctly. Several layout adjustments were made after testing on different screen sizes.

I also reviewed the code and removed unnecessary sections to improve overall organization.

20.06.2026

Today I completed the Discover Gran Canaria project and prepared it for submission.

I reviewed all project requirements and confirmed that the application includes multiple views, interactive components, image content, navigation, reusable components, and educational information.

I also prepared the project documentation, README file, and learning diary.

Final Reflection:

Throughout this project I learned how to build a complete Android application using Android Studio and Jetpack Compose. I gained practical experience with navigation, reusable components, image handling, state management, layouts, and user interface design.

The Discover Gran Canaria project allowed me to apply the concepts learned throughout the course in a realistic scenario. The biggest challenges involved navigation, layout design, and debugging, but solving these problems significantly improved my confidence as a software developer.

Overall, this project demonstrated how multiple Android development concepts can be combined into a functional and visually appealing mobile application. I now have a much stronger understanding of Android development and feel more confident creating larger applications in the future.